

Unit 1: Introduction to Operations Research (5hrs)

Introduction, History of Operations Research, Stages of Development of Operations Research, Relationship between Manager and OR Specialist, OR Tools and Techniques, Applications of Operations Research, Limitations of Operations Research

#Past Questions

1. What is Operational Research? Explain the general methods for solving OR models. –[5] [2021/2023]
2. Describe in brief, the tools and techniques of OR.-[2.5+2.5] [2024]

1. Introduction to Operations Research (OR)

Operations Research (OR) is a scientific and systematic method of solving complex decision-making problems. It involves using mathematical models, statistical analysis, and computational techniques to optimize processes and help organizations make better decisions.

The main goal of OR is to find the best possible solution for resource allocation, planning, scheduling, and other operational challenges under given constraints.

It helps in choosing the **best possible solution** among many options, especially when resources (time, money, materials, labor) are limited.

Why is OR Important?

- Improves **efficiency** and **productivity**.
- Reduces costs and waste.
- Enhances decision-making under uncertainty.
- Supports strategic planning and policy design.

Examples:

A company wants to minimize shipping costs while delivering products to customers.

- **OR Approach:**
 - Model the problem as a **transportation problem** (a type of linear program).
 - Use algorithms to determine the optimal routes and shipments.
 - Result: Cost savings and faster delivery times.
- A company wants to **maximize profit** by producing two products (say, Tables and Chairs) using limited resources like wood and labor. OR can help find **how many units** of each product to produce to get the **maximum profit**, while not exceeding the resource limits.

- A hospital using OR techniques to optimize patient flow, reduce wait times, and improve resource allocation. This could involve analyzing patient arrival rates, queuing patterns, and staffing needs to identify bottlenecks and areas for improvement.

Other Examples:

Logistics and Supply Chain: OR can be used to optimize delivery routes for couriers, minimizing fuel consumption and delivery times.

Healthcare: OR can be used to determine the minimum number of staffs needed for a healthcare facility to operate normally, considering constraints like available positions, salaries, and licenses.

Transportation: OR can be used to optimize airline routes, minimize flight delays, and improve resource allocation.

Manufacturing: OR can help a food processing plant determine the optimal number of hot dogs and hot dog buns to produce, maximizing profit while respecting resource constraints.

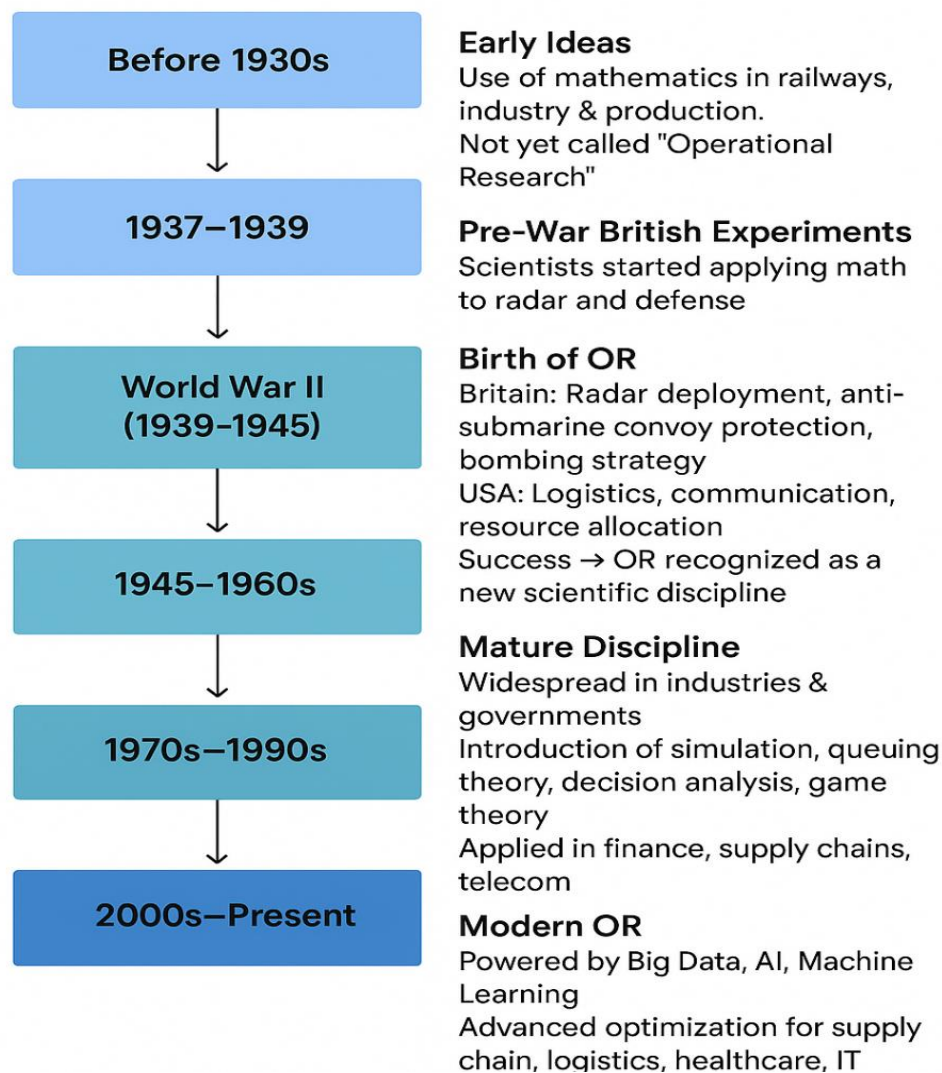
Investment: OR can be used to optimize investment portfolios, maximizing returns while managing risk.

HISTORY OF OPERATIONS RESEARCH

- OR came into being during second world war-II (1939-1945). During this war, military management of England had called up on a team of scientists to study the problems related to air and land defence of the country. The team had to come out with the plans that could make the optimal utilization of limited available resources. The research work done by this team on military operations was named as **Operational Research**, Operational Analysis, Operations Evaluation, and Systems Research and Management Science.
- After the end of world war, industrial managers got attracted towards Operational Research teams for finding solutions to their complex executive-type problems.

- The first mathematical technique called Simplex Method of Linear Programming was developed in 1947 by American mathematician George B. Dantzig.
- Operational Research Club of Britain was formed in 1948, which in 1954 became the Operational Research Society.
- The Operations Research Society in America was formed in 1952.
- The first international conference on operations research was held at Oxford University in 1957.
- In 1959 an International Federation of Operational Research Societies was formed.

- Operations Research appeared as an academic discipline for the first time in 1948 in Massachusetts Institute of Technology.
- In 1952 a curriculum leading to a master's and doctoral degree was established at the Case Institute of Technology (now Case Western Reserve University) in Cleveland.
- The first chair in operations research was created at the newly formed University of Lancaster in 1964. Similar developments have taken place in most countries in which a national operations research society exists.
- In 1949, the first Operational Research unit was established at Hyderabad which was named Regional Research Laboratory.
- At the same time an additional unit was launched in Defence Science Laboratory to fix the Stores, Purchase and Planning Problems.
- In 1953 at Calcutta, an O.R. unit was established in Indian Statistical Institute. The objective was to use O.R. techniques in National Planning and Survey.
- In 1955 Operations Research Society of India was created, which is among the first members of International Federation of Operations Research Societies.
- Today, the utilization of O.R. techniques has spread out from army to a wide range of departments at all levels.



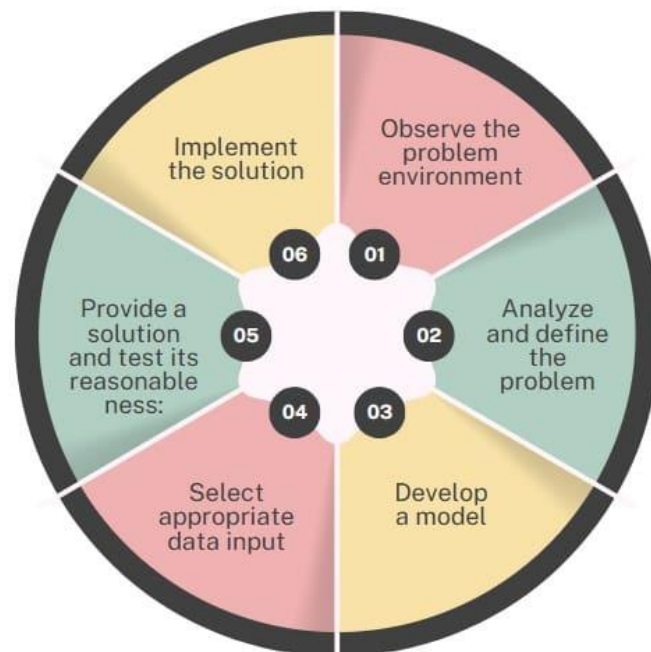
3. Stages of Development of OR

1. Observe the Problem Environment

- **What it means:** Understand the real-world situation where the problem exists.
- **Purpose:** Gather background information, identify stakeholders, and recognize constraints or limitations.
- **Example:** In a delivery company, observing how packages are sorted and shipped reveals where inefficiencies may exist.

2. Analyze and Define the Problem

- **What it means:** Clearly state what the actual problem is, including its scope, objectives, and limitations.
- **Purpose:** To ensure that everyone understands the issue the same way and agrees on what needs to be solved.



- **Example:** The problem may be "minimizing delivery time while reducing fuel costs."

3. Develop a Model

- **What it means:** Create a mathematical or logical representation of the real-world problem.
- **Purpose:** A model helps simulate the situation to test different scenarios and solutions without disrupting the real system.
- **Example:** A linear programming model for minimizing transportation costs.

4. Select the Appropriate Data Input

- **What it means:** Collect and verify the data required to use in the model.
- **Purpose:** The accuracy of the model depends heavily on accurate and relevant data.
- **Example:** Gathering data like distances, delivery times, fuel costs, and truck capacities.

5. Provide a Solution and Evaluate Its Reasonableness

- **What it means:** Solve the model using appropriate techniques and assess if the solution makes sense in the real world.
- **Purpose:** Ensures that the model gives realistic, feasible, and beneficial outcomes.
- **Example:** A solution might minimize cost, but you must check if it's practical in terms of existing resources or time.

6. Implement the Solution

- **What it means:** Put the chosen solution into action in the real-world system.
- **Purpose:** To apply the optimized or improved strategy in practice and monitor its impact.
- **Example:** Changing the delivery routes based on the optimized model.

The stages of operational research (OR) generally involve: 1. Problem Definition, which involves identifying and clarifying the problem and its objectives; 2. Model Formulation, where a mathematical or simulation model is created to represent the problem; 3. Solution Generation, where the model is solved to find potential solutions; 4. Validation, checking the accuracy and relevance of the model and its solutions; and 5. Implementation, which involves putting the chosen solution into practice and monitoring its effectiveness.

1. Problem Definition

- **Observe the environment:** Understand the system where the problem exists.
- **Define the problem:** Clearly state the issue, including its objectives, constraints, and limitations.
- **Set objectives:** Determine what the solution should achieve (e.g., increase profits, maintain stable prices).

2. Model Formulation

- **Data Collection:** Gather necessary data from various sources, ensuring its accuracy and relevance.
- **Develop a model:** Create a mathematical, logical, or simulation model that represents the problem's key variables and relationships.
- **Quantify the problem:** Express all factors and variables as numbers for the model.

3. Solution Generation

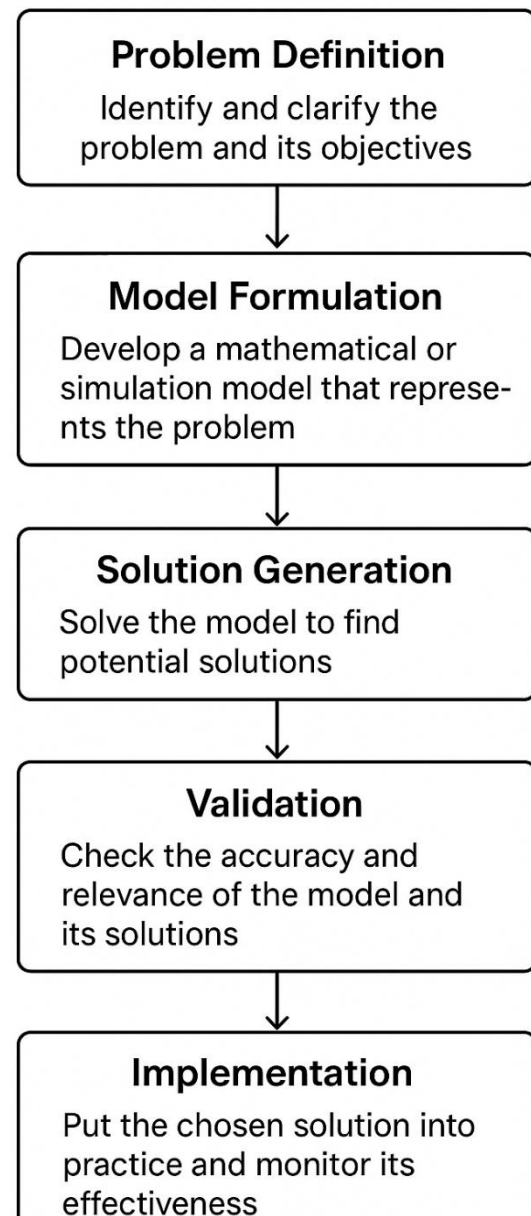
- **Derive solution(s):** Apply appropriate techniques to the model to find potential optimal solutions.
- **Analyze alternatives:** Identify and compare the various possible solutions generated by the model.

4. Validation

- **Test the model:** Verify that the model is accurate and represents the problem effectively.
- **Analyze the output:** Evaluate the chosen solution's optimality and ensure it makes practical sense.

5. Implementation and Monitoring

- **Implement the solution:** Put the best-validated solution into practice in the real-world situation.
- **Monitor and control:** Continuously track the solution's performance to ensure it delivers the desired results and make adjustments as needed.



4. Relationship Between Manager and OR Specialist

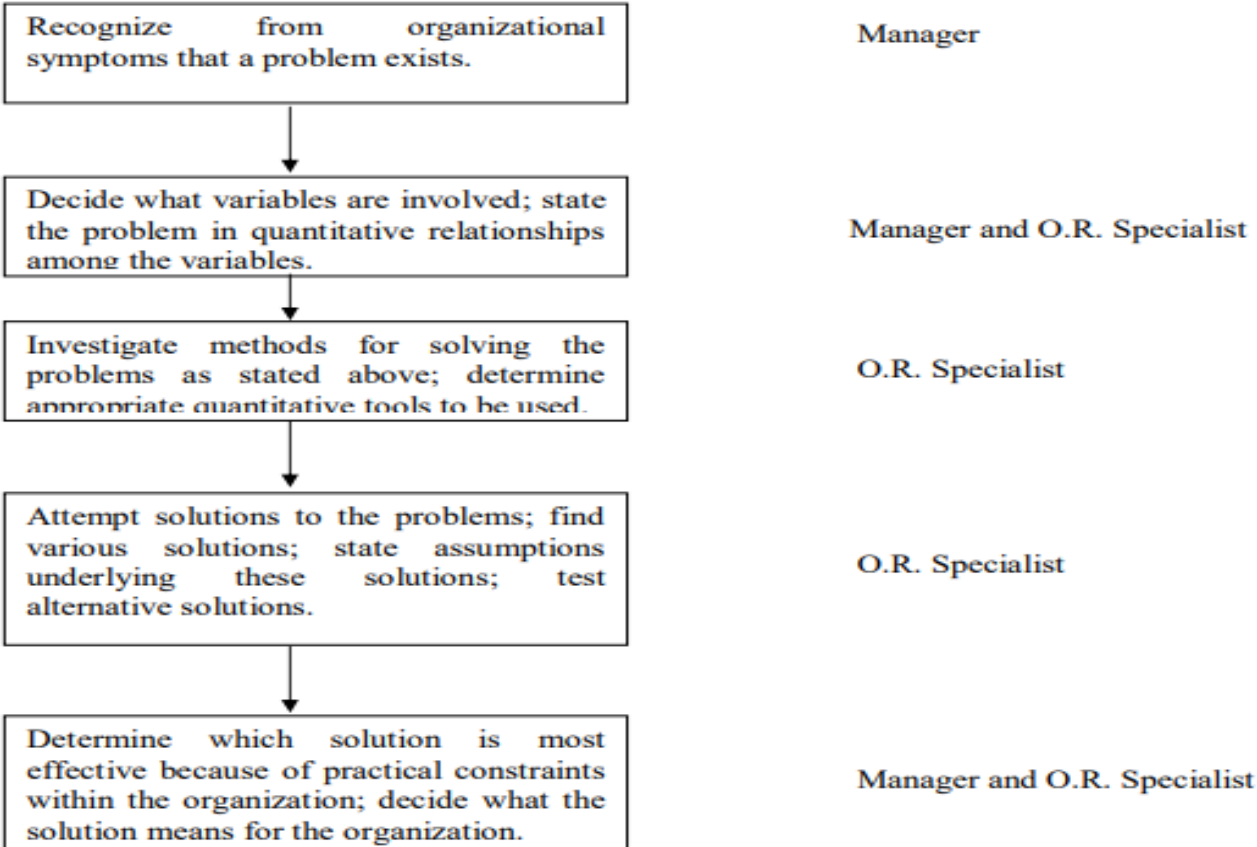
Manager	OR Specialist
Understands business needs and environment	Understands OR techniques and mathematical tools
Defines the problem in practical business terms	Translates the problem into a mathematical model

Manager	OR Specialist
Makes the final decision	Provides solution alternatives based on analysis
Applies the chosen solution in real-world	Ensures model accuracy and validity
Guides strategic direction and priorities	Develops and tests models to optimize solutions
Evaluates implementation feasibility	Suggests improvements based on quantitative analysis
Balances risks and benefits in decision-making	Evaluates sensitivity and scenario outcomes
Approves resource allocation	Supports management with insights and recommendations
Monitors outcomes post implementation	Updates models with new data for continuous improvement
Encourages collaboration among teams	Collaborates with experts and data specialists

The key responsibility of manager is decision making. The role of the O.R. specialist is to help the manager make better decisions. Figure 1-1 explains the relationship between the O.R. specialist and the manager/decision maker.

STEPS IN PROBLEM RECOGNITION,
FORMULATION AND SOLUTION

INVOLVEMENT: O.R. SPECIALIST or
MANAGER



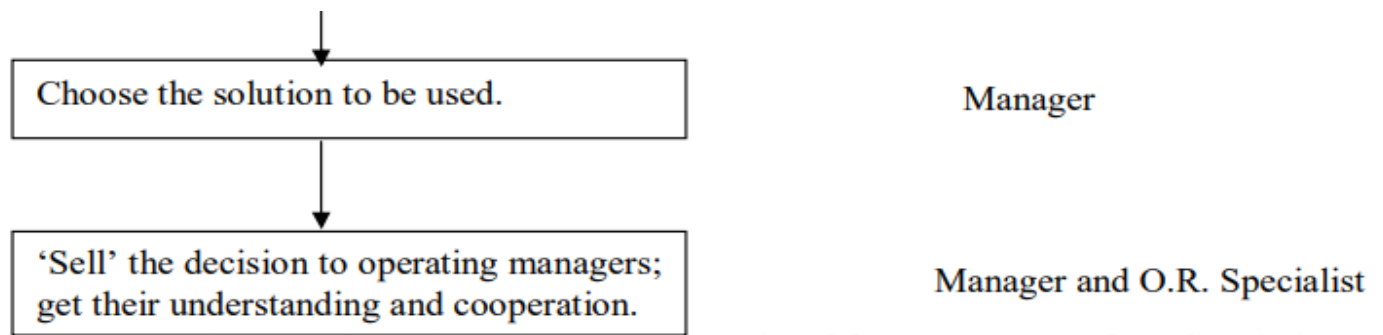


Figure 1-1 Relationship between Manager/Decision Maker and O.R. Specialists

Example: A **factory manager** wants to increase productivity. He provides data on machinery, labor, and shift times. The **OR expert** uses Linear Programming to find the best production schedule.

5. OR Tools and Techniques

(i) Linear Programming (LP)

- Used when the objective function and constraints are **linear**.
- Goal: Maximize or minimize (e.g., profit, cost).
- Methods:
 - **Graphical Method** (for 2-variable problems).
 - **Simplex Method** (for large-scale problems).

(ii) Transportation and Assignment Methods

- **Transportation Model:** Minimizes the cost of distributing a product from multiple sources to multiple destinations.
- **Assignment Model:** Allocates tasks/jobs to machines or workers in such a way that total cost/time is minimized (Hungarian Method).

(iii) Game Theory

- Used for competitive situations involving **two or more decision-makers**.
- Helps to find **optimal strategies** for players in zero-sum games.

(iv) Queuing Models

- Deals with problems of **waiting lines** (queues).
- Applied in service systems like banks, hospitals, call centers.
- Objective: Minimize waiting time and service cost.

(v) Inventory Models

- Decide the **optimum level of stock** to balance demand and supply.
- Models: Economic Order Quantity (EOQ), Probabilistic Models.

(vi) Simulation

- Used when problems are **too complex for mathematical modeling**.
- A computer is used to **simulate real-life situations** (e.g., traffic flow, factory production).

(vii) **Dynamic Programming**

- Solves **multi-stage decision problems** step by step.
- Example: Shortest path problem, project planning.

(viii) **Network Models**

- Techniques like **PERT (Program Evaluation and Review Technique)** and **CPM (Critical Path Method)** are used for project scheduling and time-cost trade-offs.

Tool/Technique	Description	Example
Linear Programming (LP)	Mathematical technique to optimize a linear objective function under constraints.	Maximize profit or minimize cost.
Transportation Model	Optimize transportation of goods from sources to destinations at minimum cost.	Determine the cheapest delivery route.
Assignment Model	Assign tasks or resources to agents to minimize cost or maximize efficiency.	Assign workers to jobs optimally.
Queuing Theory	Analyze waiting lines to reduce service time and costs.	Minimize customer wait time in banks.
Inventory Models	Manage stock levels efficiently to avoid overstocking or shortages.	Decide optimal reorder level for products.
Simulation	Create a virtual model to imitate real-world systems for testing.	Simulate hospital emergency room operations.
Game Theory	Study competitive situations to find optimal strategies.	Pricing strategy between competing firms.
Decision Theory	Helps in making decisions under uncertainty.	Choosing the best investment option.
Network Models (PERT/CPM)	Project scheduling and management to meet deadlines.	Construction project timeline planning.

6. Applications of Operations Research

1. **Business & Management**

- Optimal resource allocation (e.g., manpower, materials)
- Profit maximization and cost minimization
- Inventory and production planning

2. **Transportation & Logistics**

- Route optimization (e.g., shortest path, fuel efficiency)

- Scheduling of flights, trains, and buses
- Supply chain and warehouse management

3. Manufacturing & Industry

- Production scheduling and workflow optimization
- Quality control and process improvement
- Equipment maintenance planning

4. Healthcare

- Hospital resource allocation (beds, staff, equipment)
- Surgery scheduling
- Ambulance routing and emergency response

5. Military & Defense

- Weapon system design and deployment
- Strategy planning and logistics
- Surveillance and patrolling optimization

6. Finance & Banking

- Portfolio optimization
- Risk analysis and decision-making
- Loan and credit scoring models

7. Telecommunications

- Network design and bandwidth optimization
- Call routing and traffic analysis

8. Energy Sector

- Power grid management
- Fuel mix optimization
- Load forecasting and scheduling

9. Sports

- Team selection and game strategy
- Tournament scheduling
- Performance analysis

10. Public Sector & Government

- Urban planning and traffic control
- Disaster management
- Policy decision-making and budgeting

Field	Application	Example
Manufacturing	Production scheduling	Decide how much to produce of each product
Transportation	Route planning	Shortest path for delivery trucks
Healthcare	Staff scheduling	Assign doctors and nurses to shifts
Banking	Queue management	Minimize customer wait time at counters
Retail	Inventory control	Maintain optimal stock levels of items
Defense	Resource allocation	Optimal troop deployment

7. Limitations of Operations Research

1. Data dependency

OR models demand accurate, high-quality quantitative data. Incomplete or outdated data can mislead decisions

2. Simplifying assumptions

Models often assume linearity, fixed parameters, or ignore variability—simplifications that may not reflect real-world complexity .

3. Computational complexity

Large-scale problems involve massive calculations requiring time and resources, sometimes making real-time solutions impractical .

4. Unquantifiable factors

Qualitative elements like human behavior, culture, ethics, and managerial intuition are hard to model and often excluded .

5. Gap between analysts and users

Specialists (mathematicians/statisticians) may struggle to translate models to business users—and vice versa—hindering acceptance .

6. High cost and resource requirements

Involving expensive technology, specialized software, and expert teams—especially burdensome for smaller organizations .

7. Implementation challenges

Organizational resistance to change, cultural inertia, and difficulties deploying solutions in practice are common hurdles .

8. Model inflexibility

OR models can quickly become outdated in dynamic environments, requiring continuous updates which can be impractical .

9. Short-term focus

They tend to optimize for immediate gains, sometimes overlooking long-term strategic considerations .

10. Sensitivity and robustness issues

Small changes in input data can drastically affect outputs, complicating trust in model suggestions

Limitation	Explanation
Data Dependency	OR models need accurate data; wrong data gives wrong solutions
Complexity	Mathematical models can be difficult to understand or solve
Time & Cost	Time-consuming to build models; software may be costly
Simplifying Assumptions	Models often assume linearity or certainty, which may not reflect real situations
Resistance to Change	People may not trust or accept model-based decisions

Summary: Operations Research is like a **scientific assistant** that helps managers make better decisions using **math and logic**. It can find the **best solutions** for production, scheduling, transportation, and more—but it depends on good data and cooperation between technical experts and managers.

#Past Questions Solution

1. What is Operational Research? Explain the general methods for solving OR models. –[5]
[2021/2023]

General methods for solving Operations Research (OR) models can be broadly categorized into analytical methods, iterative methods, and simulation methods (like Monte Carlo). Analytical methods rely on mathematical formulas and equations to find exact solutions. Iterative methods involve starting with an initial solution and repeatedly improving it until an optimal or near-optimal solution is reached. Simulation methods, such as Monte Carlo, use random sampling to analyze model behavior and estimate solutions.

1. Analytical Methods:

- These methods use mathematical techniques like calculus, linear programming, and other algebraic manipulations to find precise solutions to OR models.
- They are suitable when the model can be represented by a set of equations or inequalities with known properties.

- Examples include using calculus to optimize a function or employing linear programming techniques to solve resource allocation problems.

2. Iterative Methods:

- These methods are employed when analytical solutions are difficult or impossible to obtain.
- They start with an initial guess or solution and then systematically refine it through a series of steps.
- Examples include the Simplex method for linear programming, where the algorithm moves from one feasible solution to another until an optimal one is found.

3. Simulation Methods (Monte Carlo):

- These methods rely on random sampling to simulate the behavior of a system represented by the OR model.
- They are particularly useful for complex systems with probabilistic elements.
- The Monte Carlo method involves generating random numbers to represent uncertain variables and running the simulation many times to observe the system's behavior.
- By analyzing the results of these simulations, estimates of the optimal solution or performance measures can be obtained.